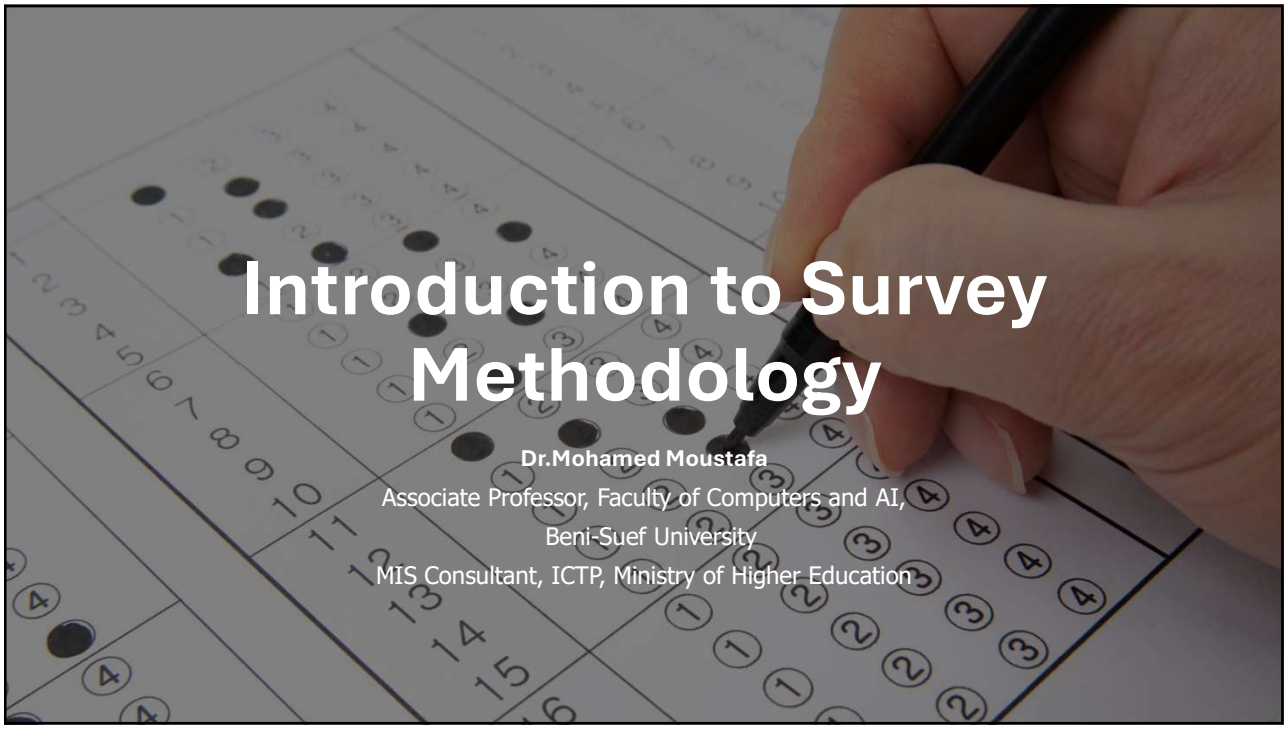

Introduction to Survey Methodology

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1

Class Rules

- You can do anything except:
  - Make noises (chatting, singing…)
- Feel free to interrupt me if you have questions .
- According to the university policy, taking attendance is needed.
  - **Important: you are required to have an 80% attendance to be able to seat for the final exam.**

2

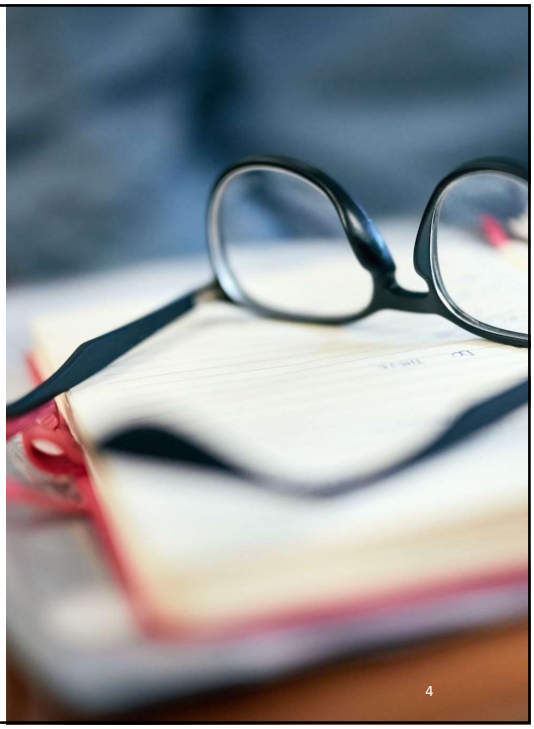
Course Assessment

- Temporary according to the situation:
  - Final exam: 50%
  - Assignment: 20%, individually
  - Project: 30%, 2-3 members per group, report and presentation are required.
- **Important: cheating and plagiarism will get no marks.**

3

A few suggestions….

- Your final grade is based on points – not on an accumulation of grades.
- You start the class with zero points and earn your way to your final grade
- If you have an issue or problem, communicate – send me an email
- If you know you're not going to meet the deadline for a quiz or assignment – email me BEFORE the deadline



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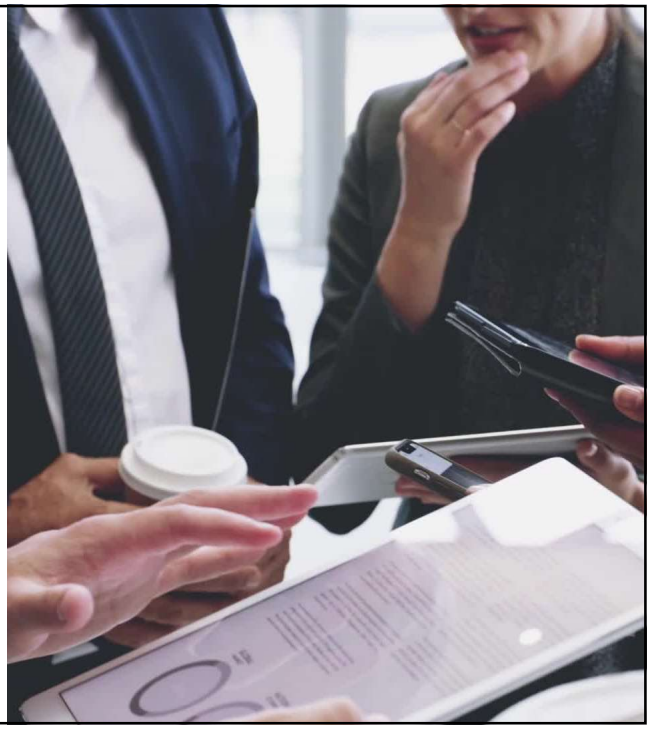
What is a Survey?

• A **survey** is a method of collecting information from a group of people to gather insights on opinions, behaviors, or characteristics.



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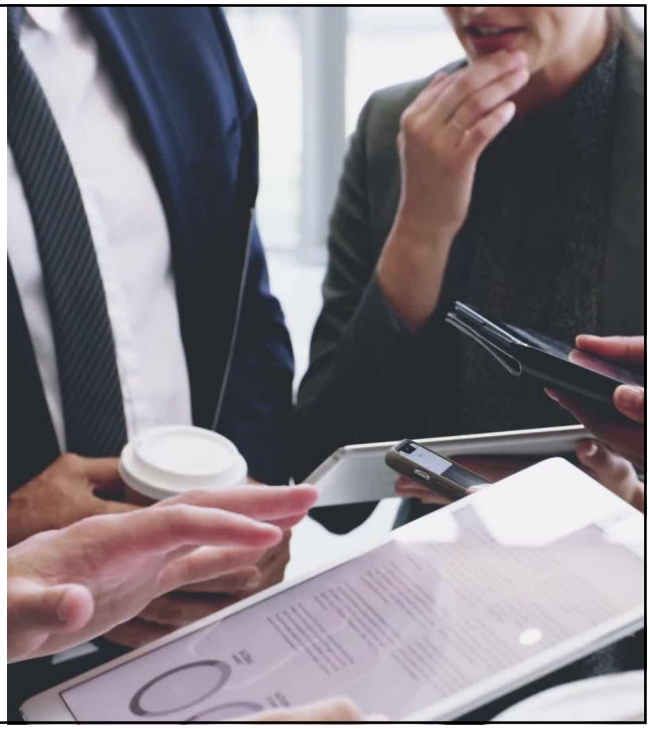
Why are surveys important?

- They help **collect large-scale data efficiently**.
- They support **decision-making** in business, government, healthcare, and academia.
- They are used in **predictive analytics** and **machine learning models**.
- **Example:**
  - A company wants to know customer satisfaction after launching a new mobile app.
  - A university wants to measure students' experiences with online learning.



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Key Components of a Survey:

Population: The entire group being studied (e.g., all customers of a brand).

Sample: A subset of the population used for data collection (e.g., 500 customers).

Questions: The structured inquiries used to collect data (e.g., "On a scale of 1-5, how satisfied are you?").

7

Types of Surveys

- Based on Data Collection Method
- Based on Time Frame
- Experimental vs. Non-Experimental Surveys

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Based on Data Collection Method

Type	Description	Example
Online Surveys	Conducted using digital tools	Google Forms, Qualtrics
Telephone Surveys	Conducted via phone calls	Customer feedback surveys
Face-to-Face Interviews	Direct interaction with respondents	Government census
Paper-Based Surveys	Physical forms/questionnaires	Election polls

9

Interviews

- Interviews help you dig through your users' knowledge base, so you can understand what they understand and think.
- One writer notes, "Interviews provide an efficient way to collect large amounts of in-depth data quickly,"
- **Benefit: By exploring someone's knowledge and needs in-depth, one-on-one, you ensure you understand the real, not just the perceived, need.**

10

Planning the Interview

- Five steps in planning the interview are
  - Reading background material
  - Establishing interview objectives
  - Deciding whom to interview
  - Preparing the interviewee
  - Deciding on question types and structure

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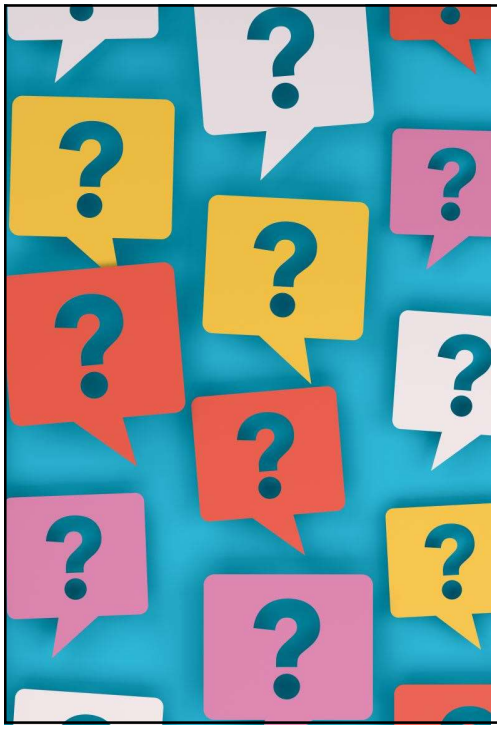

Open-Ended Questions

- Open-ended interview questions allow interviewees to respond how they wish, and to what length they wish
- Open-ended questions are appropriate when the analyst is interested in breadth and depth of reply
- Example:
  - What's your opinion of the present computer system?
  - How do you view the goals of this department?
  - What are some of common errors made in data entry in department?
  - Describe…….

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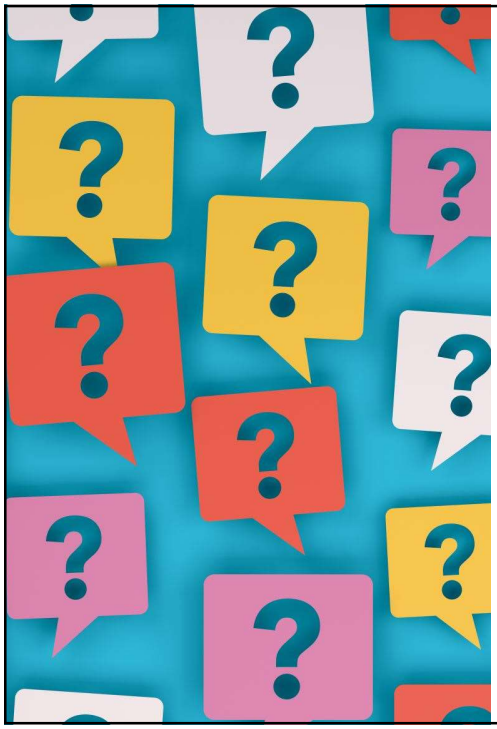

Advantages of Open-Ended Questions

- Putting the interviewee at ease
- Allowing the interviewer to pick up on the interviewee's vocabulary
  - Reflect education, values, attitudes, and beliefs
- Providing richness of detail
- Revealing avenues of further questioning that may have gone untapped
- More interesting for the interviewee
- Allows more spontaneity
- Makes phrasing easier for the interviewer
- Useful if the interviewer is unprepared

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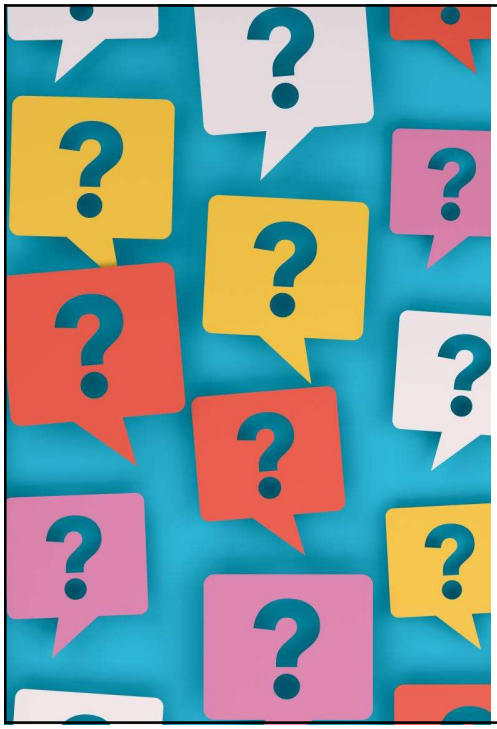

Disadvantages of Open-Ended Questions

- May result in too much irrelevant detail
- Possibly losing control of the interview
- May take too much time for useful information gained
- Potentially seeming that the interviewer is unprepared

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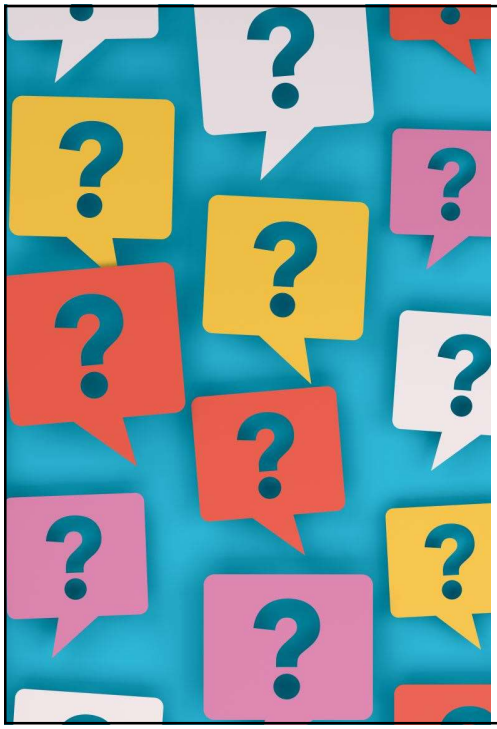

Closed Interview Questions

- Closed interview questions limit the number of possible responses
- Closed interview questions are appropriate for generating precise, reliable data which is easy to analyze
- The methodology is efficient, and it requires little skill for interviewers to administer
- Example:
  - How many report do you generate in month?
  - List your top two priorities for the marketing department?
  - Who receives this output?

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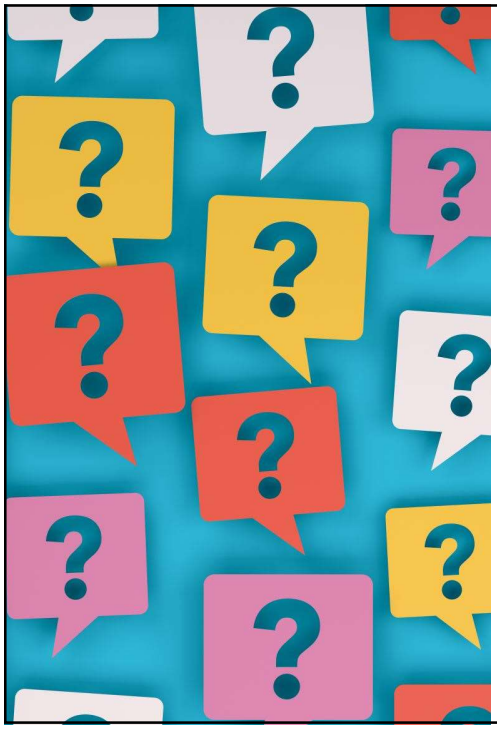

Advantages of Closed Interview Questions

- Saving interview time
- Easily comparing interviews
- Getting to the point
- Keeping control of the interview
- Covering a large area quickly
- Getting to relevant data

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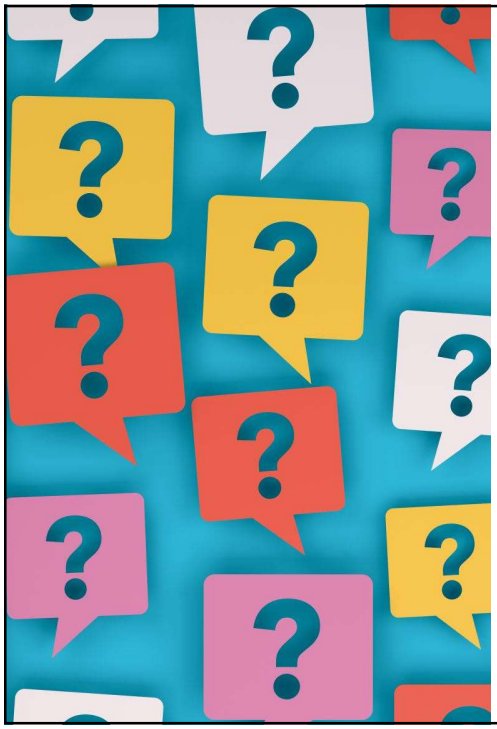

Disadvantages of Closed Interview Questions

- Boring for the interviewee
- Failure to obtain rich detail
- Missing main ideas
- Failing to build rapport between interviewer and interviewee

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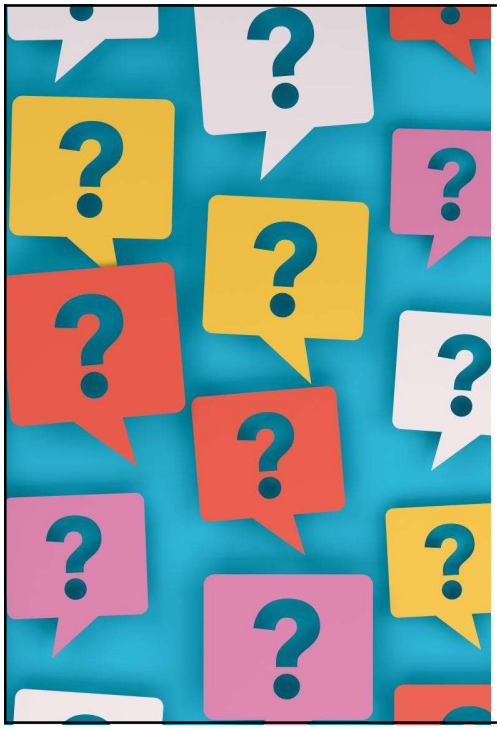

Bipolar Questions

- Bipolar questions are those that may be answered with a ‘yes’ or ‘no’ or ‘agree’ or ‘disagree’
- Bipolar questions should be used sparingly
- Example:
  - Do you agree or disagree that commerce on the web lacks security?
  - Do you want to receive a printout if your account status every month?
  - Is this form complete?

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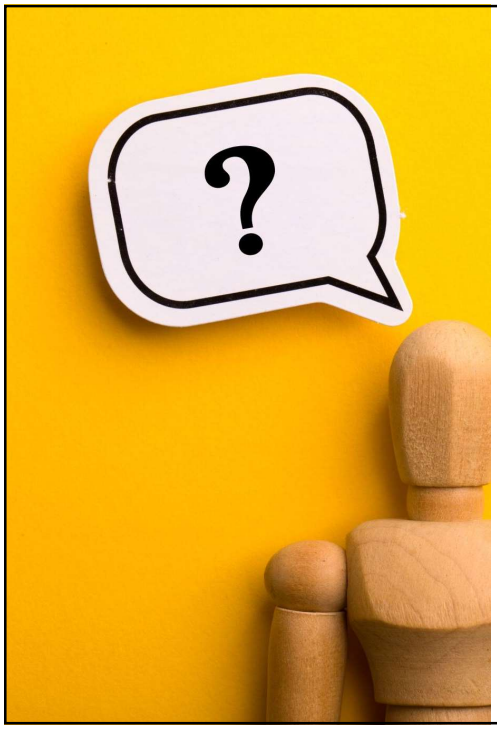

Probing Questions

- Probing questions elicit more detail about previous questions
- The purpose of probing questions is
  - To get more meaning
  - To clarify
  - To draw out and expand on the interviewee's point
- Example:
  - Why?
  - Give example of your decision-making process.
  - Tell me what happens step by step to the form the patient fills out.

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Question Pitfalls

- Avoid leading questions, those that imply an answer
- Leading questions tend to guide interviewees into responses apparently desired by the interviewer
- These questions should be avoided to reduce bias and improve reliability and validity
- Avoid double-barreled questions, asking two questions at once
- These questions should be avoided because interviewees may answer only one question, leading to difficulties in interpretation

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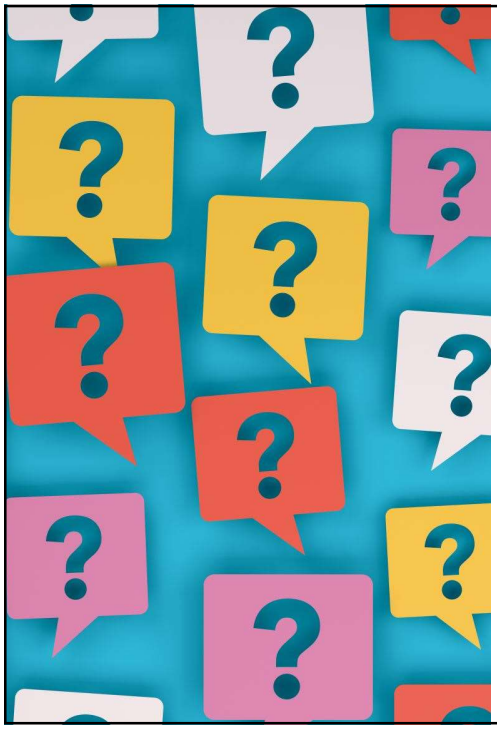

Question Sequencing

- There are three basic ways of structuring interviews:
  - Pyramid
    - Starting with closed questions and working toward open-ended questions
  - Funnel
    - Starting with open-ended questions and working toward closed questions
  - Diamond
    - Starting with closed, moving toward open-ended, and ending with closed questions

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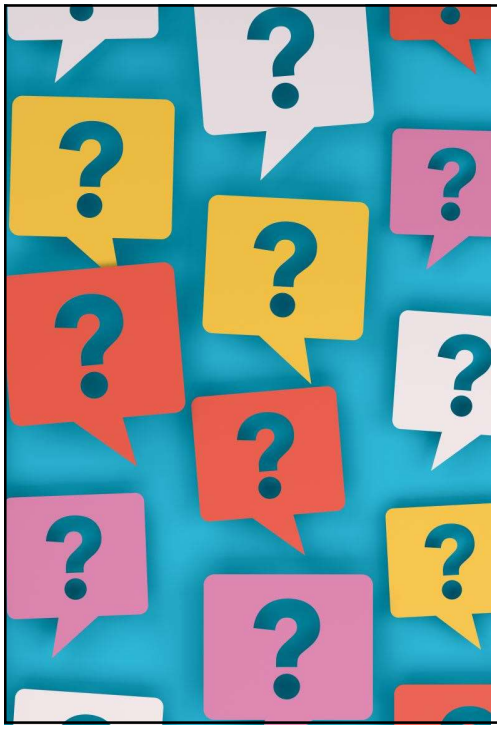
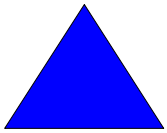


Pyramid Structure



- Begins with very detailed, often closed questions
- Expands by allowing open-ended questions and more generalized responses
- Is useful if interviewees need to be warmed up to the topic or seem reluctant to address the topic

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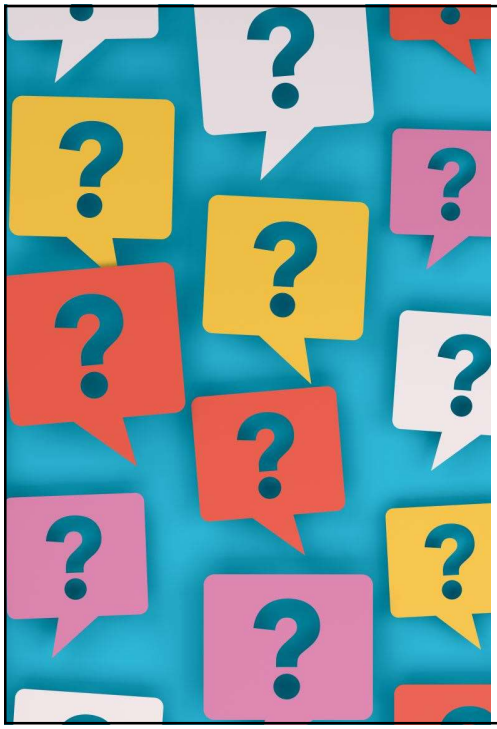

Funnel Structure

- Begins with generalized, open-ended questions
- Concludes by narrowing the possible responses using closed questions
- Provides an easy, nonthreatening way to begin an interview
- Is useful when the interviewee feels emotionally about the topic

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Interviews

For your interview to be effective, you must decide how structured or unstructured you want your interview to be.

- **Structured Interviews:**
  - These are interviews that strictly adhere to the use of an interview protocol to guide the researcher.
  - It is a more rigid interview style, in that only the questions on the interview protocol are asked.
  - As a result, there are not a lot of opportunities to probe and further explore topics that participants bring up when answering the interview questions.
  - **This method can be advantageous when researchers have a comprehensive list of interview questions**

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Interviews (contd.)

- **Semi structured interviews:**
  - These are interviews that use an interview protocol to help guide the researcher through the interview process.
  - It does maintain some structure (hence the name semi structured),
  - But it also provides the researcher with the ability to probe the participant for additional details.
  - If you decide to choose this interview method, understand that it offers a great deal of flexibility for you as a researcher.
  - You do not have to worry about needing to conduct several rounds of interviews **because your interview protocol will keep you focused on gathering all the information** that you need to answer your research question.

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Interviews (contd.)

- **Unstructured interviews:**
  - These are interviews that take place with few, if any, interview questions.
  - They often progress in the manner a normal conversation would, however it concerns the research topic under review.
  - It is a relatively formless interview style that researchers use to establish rapport and comfort with the participant and is extremely helpful when researchers are discussing sensitive topics.
  - If you select this interview style, just keep in mind that you **may have to conduct several rounds of interviews** with your participants in order to gather all the information you need.
  - Since you do not use a standard interview protocol, sometimes participant's narratives maneuver the conversation away from other aspects of the research topic you want to explore;

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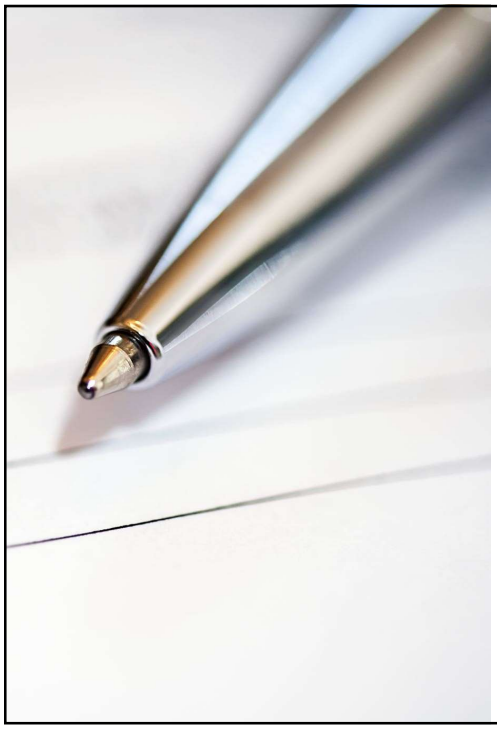

Interview Report

- Write as soon as possible after the interview
- Provide an initial summary, then more detail
- Review the report with the respondent

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Questionnaires

- Questionnaires are useful in gathering information from key organization members about:
  - Attitudes: what people in the organization say they want.
  - Beliefs: what people think is true.
  - Behaviors: what organizational members do.
  - Characteristics: properties of people or things.

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Surveys /Questionnaires

- The ideal approach to this technique is by making a basic Google Form and offering it to the correct individuals, and whenever required, determining a due date.
- You must know what you are attempting to accomplish precisely with the study, and the questions must not to be uncertain.
- Misunderstanding of inquiries can prompt useless and pointless answers.

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Questionnaire Language

- **Simple –** Use the language of the respondents whenever possible.
- **Specific –** work at being specific rather then vague in wording.
- **Short –** keep questions short
- **Not patronizing –** do not talk down to participants through low-level language choices.
- **Free of bias –** also means avoiding objectionable questions.
- **Measurement Scales:** Scaling is the process of assigning numbers to an attribute for the purpose of measuring that attribute or characteristic.
- **Reliability** of scales refers to consistency in response
- **Validity** is the degree to which the question measures what the analyst intends to measure.

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The format for Questionnaires

- **Fixed Format:**
  - Fixed format surveys consist of questions that need a variety of predefined responses from people.
  - Respondents have to choose an answer from a series of answers provided.
  - A reply from this format of the questionnaire is a lot simpler to interpret.
  - In any case, then again, it is increasingly latent; respondents can't give their answers or opinion other than presented in the survey.
- **Free Format:**
  - Free format surveys will enable users to answer openly for each inquiry.
  - A question is proposed, and the respondent enters the appropriate response in the space given after the query.

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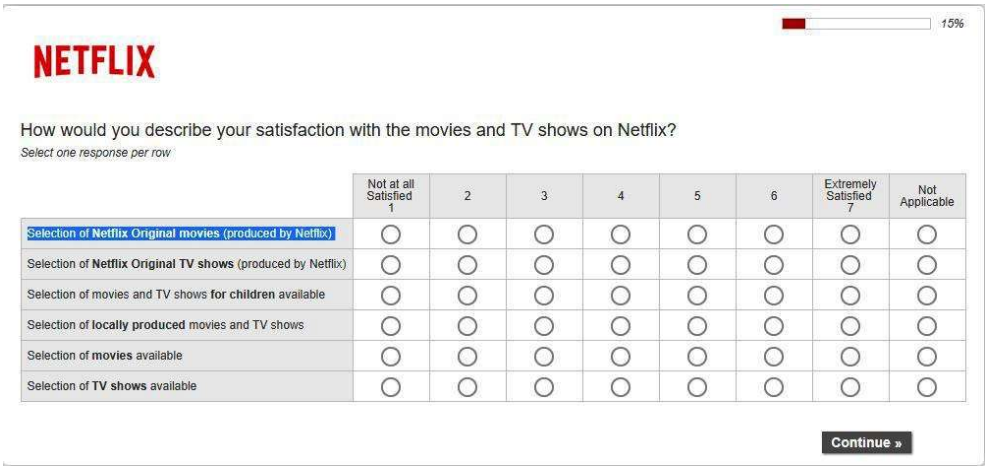
Fixed Format



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NETFLIX

How would you describe your satisfaction with the movies and TV shows on Netflix?

Select one response per row

	Not at all Satisfied 1	2	3	4	5	6	Extremely Satisfied 7	Not Applicable
Selection of **Netflix Original movies** (produced by Netflix)	○	○	○	○	○	○	○	○
Selection of **Netflix Original TV shows** (produced by Netflix)	○	○	○	○	○	○	○	○
Selection of movies and TV shows **for children** available	○	○	○	○	○	○	○	○
Selection of **locally produced** movies and TV shows	○	○	○	○	○	○	○	○
Selection of **movies** available	○	○	○	○	○	○	○	○
Selection of **TV shows** available	○	○	○	○	○	○	○	○

Continue »

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Free Format

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Observation

- Observing the decision maker; and the decision maker's physical environment; and their interaction with their physical, ergonomic environment are important unobtrusive methods.
- Observation provides insight on what organizational members do.
- Help confirm what has been found through other methods.

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Based on Data Collection Method

•Which survey method do you think is the **most reliable**?
•What are some **problems with online surveys**?

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Note

- The most reliable method depends on the study.
- Face-to-face surveys offer higher accuracy, but online surveys work well with proper design and safeguards.

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Example 1: 2016 U.S. Election Polling Errors

- **Survey Type:** Online & Telephone Political Polls
  Issue: Non-representative Sampling & Self-Selection Bias
- **What happened?**
  - Many pre-election polls **predicted Hillary Clinton would win** by a large margin.
  - However, they **oversampled college-educated voters** and **undersampled rural voters**, leading to **inaccurate predictions**.
- **Why was this a problem?**
  - **Self-Selection Bias** – Many respondents were from urban areas, ignoring key rural voters.
  - **Non-Representative Sample** – The sample didn't accurately reflect the actual voter demographics.

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Example 2: Apple's iPhone Customer Satisfaction Survey (2018)

- **Survey Type:** Online Customer Satisfaction Survey
  Issue: Leading & Loaded Questions
- **What happened?**
  - Apple sent customers a survey asking:
    "How much do you love your iPhone?"
  - This **forced a positive response** because it assumed people already **liked** their iPhones.
- **Why was this a problem?**
  - **Leading Question Bias** – The wording pressured respondents to be positive.
  - **Lack of Neutral Option** – There was no choice for people who **disliked** their iPhone.

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